IORA Project Report

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1 Problem 1: Single-Nurse Routing and Scheduling

1.1 Problem modelling

We study the daily plan of a single nurse who departs from and returns to the medical centre. The patient set is denoted by $\mathcal{P} = \{1, \ldots, P\}$ and the node set is $\mathcal{N} = \{0\} \cup \mathcal{P} \cup \{P+1\}$, where node 0 is the departure depot and node $P+1$ is a dummy sink that represents the return to the depot. Each patient $i \in \mathcal{P}$ has a service duration d_i, a time window $[S_i, F_i]$ within which service must start and finish, and pairwise travel times T_{ij} are known for all depot–patient and patient–patient pairs. The earliest departure from the depot is restricted by L, and the nurse must return no later than A.

To simplify the routing, the travel matrix is extended by copying the travel time from every patient back to the depot onto the arc that connects the patient to the sink node. The model enforces that exactly one arc leaves the depot, exactly one arc enters the sink, and each patient has one predecessor and one successor, thereby yielding a single Hamiltonian tour that visits all patients once. Service start times are tracked explicitly and, combined with service durations and travel times, they implicitly remove subtours when linked through precedence constraints.

1.2 Mixed-integer linear formulation

Define binary routing variables x_{ij} that indicate whether arc (i, j) is traversed, and continuous variables s_i that denote the start of service at node i. The set of feasible arcs is $\mathcal{A} = \{(i, j) \mid i \in \{0, \ldots, P\},\ j \in \{1, \ldots, P+1\},\ i \neq j,\ (i, j) \neq (0, P+1)\}$. The resulting mixed-integer linear program is

$$\min \quad s_{P+1} - s_0 \tag{1}$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{P}} x_{0j} = 1, \tag{2}$$

$$\sum_{i \in \{0\} \cup \mathcal{P}} x_{i,P+1} = 1, \tag{3}$$

$$\sum_{j \in \mathcal{P} \cup \{P+1\} \setminus \{i\}} x_{ij} = 1 \qquad \forall i \in \mathcal{P}, \tag{4}$$

$$\sum_{i \in \{0\} \cup \mathcal{P} \setminus \{j\}} x_{ij} = 1 \qquad \forall j \in \mathcal{P}, \tag{5}$$

$$s_0 \geq L, \qquad s_{P+1} \leq A, \tag{6}$$

$$S_i \leq s_i \leq F_i - d_i \qquad \forall i \in \mathcal{P}, \tag{7}$$

$$s_j \geq s_i + d_i + T_{ij} - M(1 - x_{ij}) \qquad \forall (i,j) \in \mathcal{A}, \tag{8}$$
$$x_{ij} \in \{0,1\} \qquad \forall (i,j) \in \mathcal{A}, \tag{9}$$
$$s_i \geq 0 \qquad \forall i \in \mathcal{N}. \tag{10}$$

Constraint (2) forces exactly one departure from the depot and (3) forces exactly one return arc into the sink. Constraints (4)–(5) ensure that each patient is visited once. Inequalities (6)–(7) enforce depot and patient time windows, while (8) is a big-M precedence constraint linking the service start times along active arcs, effectively eliminating subtours thanks to strictly increasing times. The constant M is chosen as

$$M = A - L + \max_{i,j \in \{0,\ldots,P\}} T_{ij} + \sum_{i \in \mathcal{P}} d_i,$$

which is a safe upper bound on the maximum possible time difference between two consecutive service starts. The objective (1) minimises the nurse's total working time, i.e., the elapsed time between departure and return.

1.3 Solution approach

Algorithm 1 summarises the implementation in `p1.py`. The procedure constructs the extended travel matrix, builds the CBC model, and post-processes the resulting solution to recover the nurse's route and timeline segments.

Algorithm 1 MILP solution for the single-nurse routing problem

Require: Instance data $(\mathcal{P}, d, S, F, L, A, T)$
Ensure: Optimal (or best feasible) route, start times, and activity segments
1: Extend T to include the sink node by copying depot return times, obtaining T^{ext}
2: Create a CBC solver and initialise binary variables x_{ij} for $(i,j) \in \mathcal{A}$ and continuous start times s_i
3: Add depot departure/arrival constraints (2)–(3) and patient flow constraints (4)–(5)
4: Enforce time-window bounds (6)–(7) and big-M precedence constraints (8)
5: Set the objective to minimise $s_{P+1} - s_0$ as in (1) and call the solver
6: **if** CBC returns a feasible solution **then**
7: Follow active arcs (i,j) with $x_{ij} = 1$ to build the route from depot to sink
8: Record start times s_i and derive travel, wait, and service segments for visualisation
9: **else**
10: Report that the instance is infeasible for the given data
11: **end if**

1.4 Computational results

The model is implemented in `p1.py` using the CBC solver distributed with OR-Tools. Two benchmark instances are analysed separately: a small case with $P = 5$ patients and a larger case with $P = 10$ patients. In both runs, the solver identified the optimal integer solution without any integrality gap.

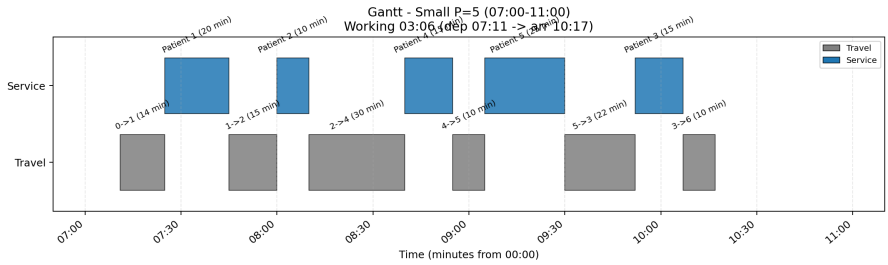


Figure 1: Gantt chart of the optimal schedule for the small instance.

Metric	Value	Notes
Route sequence	0-1-2-4-5-3-6	Depot nodes 0 and 6
Working time	186 min	07:11–10:17
Waiting time	0 min	No idle periods

Table 1: Key indicators for the small instance.

1.4.1 Small instance ($P = 5$)

The nurse departs at 07:11, services patients $(1, 2, 4, 5, 3)$ in order, and returns at 10:17. Because every arrival coincides with the opening of the patient's time window, the schedule operates without any waiting, and the Gantt chart in Figure 1 shows back-to-back service blocks.

1.4.2 Large instance ($P = 10$)

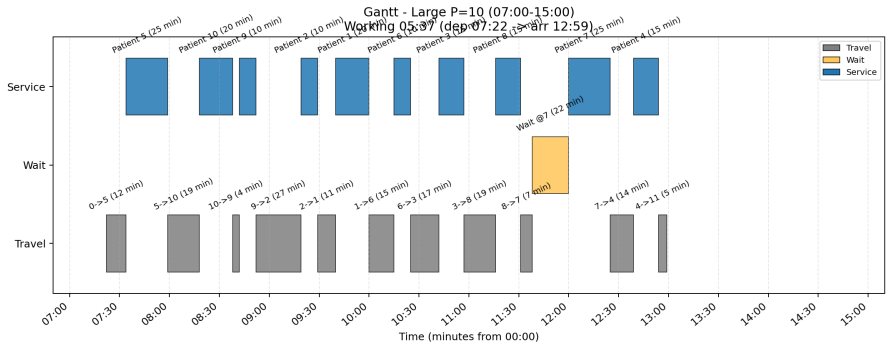


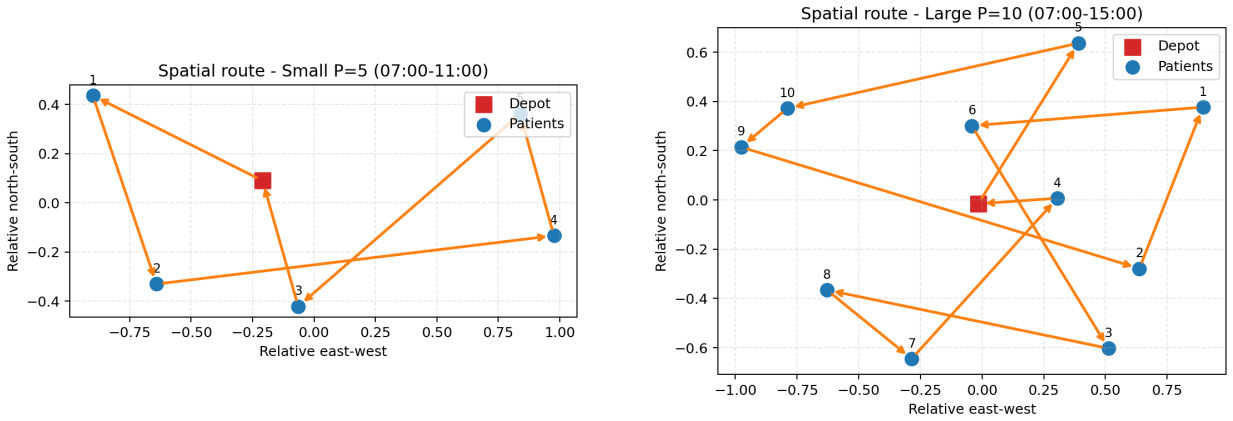
Figure 2: Gantt chart of the optimal schedule for the large instance.

Metric	Value	Notes
Route sequence	0-5-10-9-2-1-6-3-8-7-4-11	Depot nodes 0 and 11
Working time	337 min	07:22–12:59
Waiting time	22 min	Single idle period before patient 7

Table 2: Key indicators for the large instance.

The ten-patient scenario requires a longer shift: the nurse departs at 07:22 and completes the

sequence $(5, 10, 9, 2, 1, 6, 3, 8, 7, 4)$, returning just before 13:00. As highlighted in Figure 2, a single waiting period of approximately 22 minutes is necessary before visiting patient 7 whose time window opens at noon; elsewhere the nurse moves directly from one patient to the next.



(a) Small instance ($P = 5$)

(b) Large instance ($P = 10$)

Figure 3: Spatial routes of the optimal solutions.

2 Problem 2: Multi-Nurse Routing and Scheduling

2.1 Problem modelling

Problem 2 extends the home-care routing framework to a team of N nurses that must collectively cover a set of patients $\mathcal{P} = \{1, \ldots, P\}$. All nurses depart from the shared depot (node 0) and return to the same location, represented by the sink node $P + 1$. Travel times T_{ij}, service durations d_i, and patient time windows $[S_i, F_i]$ are given, as well as global depot bounds $[L, A]$ on departures and returns. Each nurse may visit any patient, but every patient must be served exactly once. Unused nurses are allowed to take the direct arc from the depot to the sink, which corresponds to staying idle for the day.

2.2 Mixed-integer linear formulation

Let $\mathcal{K} = \{1, \ldots, N\}$ be the nurse set and $\mathcal{A} = \{(i, j) \mid i \in \{0, \ldots, P\},\ j \in \{1, \ldots, P+1\},\ i \neq j\}$ the arc set built on the extended graph with sink $P + 1$. The model introduces binary variables x_{kij} indicating whether nurse k traverses arc (i, j), continuous start times s_i for patients, and depot departure/return times dep_k and arr_k for each nurse. The MILP reads

$$\sum_{j:(0,j)\in\mathcal{A}} x_{k0j} = 1 \qquad \forall k \in \mathcal{K}, \tag{11}$$

$$\sum_{i:(i,P+1)\in\mathcal{A}} x_{ki,P+1} = 1 \qquad \forall k \in \mathcal{K}, \tag{12}$$

$$\sum_{i:(i,v)\in\mathcal{A}} x_{kiv} = \sum_{j:(v,j)\in\mathcal{A}} x_{kvj} \qquad \forall k \in \mathcal{K},\ v \in \mathcal{P}, \tag{13}$$

$$\sum_{k\in\mathcal{K}} \sum_{i:(i,v)\in\mathcal{A}} x_{kiv} = 1 \qquad \forall v \in \mathcal{P}, \tag{14}$$

$$\sum_{k\in\mathcal{K}} \sum_{j:(v,j)\in\mathcal{A}} x_{kvj} = 1 \qquad \forall v \in \mathcal{P}, \tag{15}$$

$$L \le \mathrm{dep}_k \le A \qquad \forall k \in \mathcal{K}, \tag{16}$$

$$L \le \mathrm{arr}_k \le A \qquad \forall k \in \mathcal{K}, \tag{17}$$

$$S_i \le s_i \le F_i - d_i \qquad \forall i \in \mathcal{P}, \tag{18}$$

$$\mathrm{arr}_k \ge \mathrm{dep}_k \qquad \forall k \in \mathcal{K}, \tag{19}$$

$$s_j \ge \mathrm{dep}_k + T_{0j} - M\big(1 - x_{k0j}\big) \qquad \forall k \in \mathcal{K},\ j \in \mathcal{P}, \tag{20}$$

$$\mathrm{arr}_k \ge s_i + d_i + T_{i0} - M\big(1 - x_{ki,P+1}\big) \qquad \forall k \in \mathcal{K},\ i \in \mathcal{P}, \tag{21}$$

$$s_j \ge s_i + d_i + T_{ij} - M\big(1 - x_{kij}\big) \qquad \forall k \in \mathcal{K},\ i, j \in \mathcal{P},\ i \ne j. \tag{22}$$

Constraints (11)–(12) force each nurse to leave and return exactly once, enabling the possibility of an empty tour via $x_{k0,P+1} = 1$. Constraint (13) enforces flow conservation at patient nodes per nurse, while (14)–(15) guarantee that each patient receives one predecessor and one successor across the team. Bounds (16)–(18) impose the depot and patient time windows, and (19) ties departure and return times. The big-M precedence constraints (20)–(22) ensure temporal feasibility along every active arc and eliminate subtours through strictly increasing service times. A safe constant is

$$M = A - L + \max_{(i,j)} T_{ij} + \sum_{i\in\mathcal{P}} d_i,$$

identical to Problem 1. The objective minimises the total team working time:

$$\min \sum_{k\in\mathcal{K}} \left(\mathrm{arr}_k - \mathrm{dep}_k\right). \tag{23}$$

2.3 Solution approach

Algorithm 2 summarises the implementation in `p2.py`. The code supports both the CBC solver and the CP-SAT engine; the experiments below use CP-SAT, which reaches high-quality solutions within the assigned five-minute limit.

Algorithm 2 MILP/CP-SAT workflow for the multi-nurse problem

Require: Instance $(\mathcal{K}, \mathcal{P}, d, S, F, L, A, T)$, solver choice
Ensure: Nurse routes, start times, and activity segments
1: Build the extended travel matrix T^{ext} with sink $P+1$ and arc set $\mathcal{A}$
2: Create binary variables x_{kij}, start times s_i, and depot times dep_k, arr_k
3: Add constraints (11)–(22); set the objective (23)
4: Solve with a 300 s time limit; keep the best feasible solution
5: **if** a solution is found **then**
6: For each nurse, follow arcs with $x_{kij} = 1$ to reconstruct the route
7: Recover departure, arrival, and service times, and assemble travel/wait/service segments
8: **else** report infeasibility

2.4 Computational results

We evaluate two benchmark instances: a ten-patient morning horizon and a larger twenty-patient day horizon, both staffed by two nurses. CP-SAT is used in both cases.

2.4.1 Small instance ($N = 2$, $P = 10$)

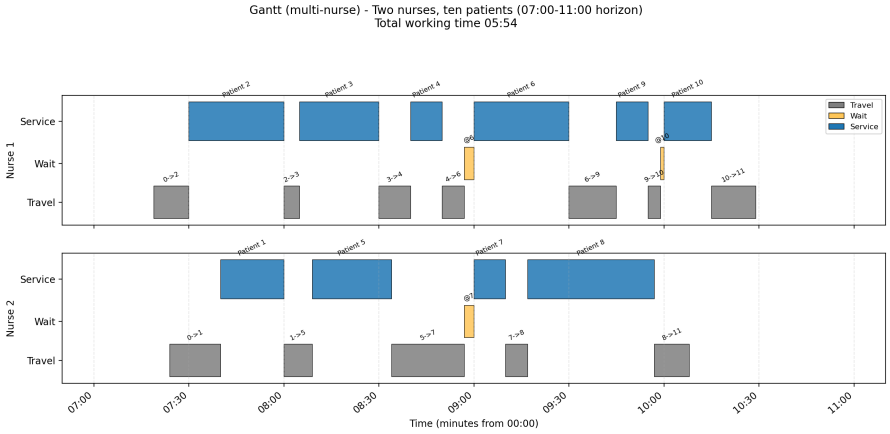


Figure 4: Gantt chart of the optimal schedule for the two-nurse, ten-patient instance.

Metric	Nurse 1	Nurse 2
Route sequence	0-2-3-7-8-11	0-1-5-4-6-9-10-11
Departure–return	07:19–10:08	07:24–10:29
Working time	169 min	185 min
Waiting time	4 min	3 min
Total working time	354 min (05:54)	

Table 3: Key indicators for the small multi-nurse instance.

CP-SAT certifies optimality in a fraction of a second. The work is balanced across the two nurses (Figure 4), each accumulating under three hours. Only seven minutes of waiting are required in total, primarily when synchronising with mid-morning time windows.

2.4.2 Large instance ($N = 2$, $P = 20$)

Metric	Nurse 1	Nurse 2
Route sequence	0-1-6-7-2-3-4-9-17-15-21	0-11-13-12-20-14-8-5-10-19-18-16-21
Departure–return	07:59–14:30	07:11–14:48
Working time	391 min	457 min
Waiting time	2 min	0 min
Total working time	848 min (14:08)	

Table 4: Key indicators for the large multi-nurse instance (best CP-SAT solution within 300 s).

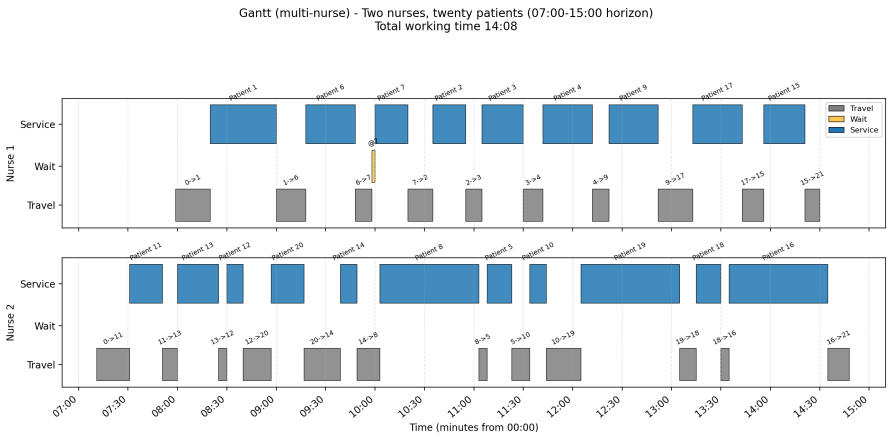


Figure 5: Gantt chart of the best solution found for the two-nurse, twenty-patient instance.

The solver hits the five-minute limit and reports a high-quality feasible solution with a total workload of 848 minutes. Nurse 2 shoulders the longer shift owing to several long services (patients 8, 19, and 16), whereas Nurse 1 covers the tighter morning windows. Only two minutes of waiting are incurred, indicating that the model effectively sequences patients around the heterogeneous time windows.